

- 2** 10 cm³ of a gaseous hydrocarbon, C_xH_y, was exploded with an excess of oxygen. There was a contraction of 40 cm³. When the products were treated with aqueous sodium hydroxide, there was a further contraction of 50 cm³. All gas volumes were measured at room temperature and pressure.

What is the molecular formula of the hydrocarbon?

- A** C₄H₈
B C₄H₁₀
C C₅H₁₀
D C₅H₁₂